Priced In or Memorized?

Parts 7–8: Baselines, Model Comparisons and Financial Evaluation

Status report: what is built, what is missing, how to continue

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In one paragraph. The code for tasks 7 and 8 is written, documented and tested (29 tests pass). It reads the outputs of parts 3–4, 5 and 6 directly, so new upstream data flows in by rerunning one command, with no code changes. With today's inputs we have: (i) the full surprise benchmark on 46,883 events (validation rank IC 0.31, $t = 22$; beat−miss reaction gap 4.5%), stable across sectors, sizes, years and sessions; (ii) the benchmark regression with controls, *including book-to-market* (Compustat, 94.6% coverage), which is the bar text must clear; (iii) a diagnostic of the day −1 placebo, which the surprise fails mildly in every subset, a pattern that looks like pre-announcement drift rather than a clock error; (iv) an exploratory Loughran–McDonald pilot on the 115 texts that exist, pointing the right way but far too small to test; (v) the error profile of the surprise (41% of beats fall) with a worked example; (vi) power with the actual sample sizes: *the pre-registered 0.02 incremental-IC bar is at or below the minimum detectable effect even in the holdout.* **The bottleneck is upstream**: release texts beyond 2006 (part 5), model scores (part 6), and the holdout freeze. No text-beyond-surprise result is possible until they arrive.

Contents

1 What parts 7 and 8 are for **2**

2 How it plugs into the rest of the project **3**

3 Design in brief **3**

4 What is built: task 7 **4**

5 What is built: task 8 **4**

6 Results so far **5**

6.1 Real data: the panel . . . 5
6.2 Real data: the surprise benchmark (validation, 2017–19) . . . 5
6.3 Real data: the benchmark regression . . . 5
6.4 Real data: beat, meet, miss . . . 6
6.5 Real data: is the benchmark stable? . . . 6
6.6 Real data: the day −1 placebo . . . 7
6.7 Real data: Loughran–McDonald pilot (exploratory) . . . 7
6.8 Real data: one figure . . . 8
6.9 Real data: where the surprise gets it wrong . . . 8
6.10 Sample sizes and power . . . 8
6.11 Synthetic data: the pipeline works . . . 9

7 What is missing **10**
7.1 Upstream inputs (blocking) 10
7.2 Decisions to write down before the freeze 10
7.3 Extensions not yet built (own work, optional) 10

8 How to continue **11**

A File map **12**

1 What parts 7 and 8 are for

The project asks whether the text of an earnings release tells us anything about the stock's abnormal return that the earnings surprise does not, and whether language models do this better than simple text models, or only appear to because they memorised outcomes.

- **Task 7** builds the simple benchmarks a language model has to beat (surprise only, Loughran–McDonald tone, TF-IDF) and compares every model on the same sample, with the same metrics, windows and chronological splits.
- **Task 8** merges model scores with the event-level financial data and runs the tests: regressions controlling for the surprise and firm characteristics, a 2×2 table (surprise sign $\times$ text sign), robustness checks, and the 2022 rate shock.

The four claims the design is built to test, in increasing order of strength:

	Claim	Test
A	Text is related to returns	Rank IC, long-short spread
B	Text adds information *beyond the surprise* (the core claim)	Text coefficient with SUE deciles; combined vs surprise-only out of sample (Clark–West); 2×2 table
C	LLMs beat simple text models	Same comparison: LLM vs Loughran–McDonald vs TF-IDF
D	It is real, not an artefact	Robustness grid, day -1 placebo, one run on the frozen holdout

2 How it plugs into the rest of the project

Part	What it provides	How 7–8 reads it
3–4	Event table: release clock, point-in-time consensus, SUE, abnormal returns (FF4 and market-adjusted), splits	`3_4_surprise_returns/data/events_final.parquet`; its `common.py` and `returns.py` are imported, not copied
5	Cleaned release texts; frozen holdout ids	`5_data cleaning/data/clean/clean_*.parquet`; `holdout_ids.txt`
6	Model scores (FinBERT, ChronoGPT, Llama 3.1 8B)	`5_data cleaning/cache/model_outputs.sqlite`, read-only; each `(model, prompt, variant)` becomes a signal `llm__<model>__<prompt>__<variant>`
9	Memorisation tests	Uses the Llama-cutoff split and the rate-shock regimes produced here
10	Report	Every table is saved as `.csv`, `.md` and `.tex`

Column contracts are checked at load time (`io_inputs.py`); a missing column fails loudly rather than producing a wrong number. Full contracts are in `docs/INTERFACES.md`.

3 Design in brief

The rationale is in `docs/METHODOLOGY.md`; the essentials are:

1. **A strong surprise benchmark.** SUE enters as decile dummies because the return response is S-shaped. A linear benchmark would let text models win by capturing that curve.
2. **Every model graded the same way.** One panel of events. Each signal is converted to its within-quarter rank in $[-0.5, 0.5]$, so a 1–5 LLM grade, a dictionary tone and a TF-IDF prediction are comparable. Comparisons use *common support*: only events every model scored.
3. **Nothing sees the future.** Trained models are fitted on earlier years only (expanding window), with a 100-day embargo so a training label's 60-day return never overlaps the test period. Tuning uses the development years only. The holdout (2021–25) needs two keys to open. A day -1 placebo catches timing errors.
4. **Inference respects clustering.** Releases cluster in quarterly seasons that share market news, so statistics are computed within season and seasons are the independent units (Newey–West, 2 lags). Regressions use two-way clustering by firm and season, with Fama–MacBeth as a check. Validation has only 12 seasons: below $|t| = 3$ we say "suggestive".

Split	Years	Role	Status
dev	2006–2016	Fit and tune	In use
validation	2017–2019	Comparison set	In use
excluded	2020	COVID year, never used	–
holdout	2021–2025	Final test, one run	Locked until freeze

4 What is built: task 7

Requirement	Implementation	Status
Surprise-only baseline	`baseline_surprise.py`: SUE rank, linear and decile models, cross-fitted	**Done** (real data)
Loughran–McDonald baseline	`baseline_lm.py`: 7 categories, negation of positive words within 3 words, signals `lm_tone`, `lm_net`, `lm_negshare`, `lm_unc`; counts cached by text hash; dictionary 1993–2025 (SRAF, March 2026)	**Done**; runs on 115 texts, **Waiting** texts
TF-IDF baseline	`baseline_tfidf.py`: uni/bigrams, numbers masked, firm filter (≥ 20 firms per term), ridge and logistic, tuned on dev only	**Done** (code); **Waiting** texts
Language models on same sample	`io_inputs.load_model_scores`: reads part 6's cache, `raw` and `mask` variants	**Done** (plumbing); **Waiting** scores
Surprise-only vs text-only vs combined	`compare.py`: out-of-sample R^2, gain over surprise-only, Clark–West with t_{S-1} p-values	**Done**
Consistent metrics, windows, splits	`metrics.py`, `crossfit.py`: same IC, long-short, hit rate, AUC, five windows, one split scheme	**Done**
Complex vs simple models	Combined-model rows for every signal in one table	**Done**; needs LLM scores

5 What is built: task 8

Requirement	Implementation	Status
Merge scores with financial data	`panel.py` + `pipeline.py`: one row per event, signals aligned by event	**Done**
Signal–return relationship	IC and decile long-short per signal and window	**Done**
Regressions with controls	`regressions.py`: ladder (1)–(5), controls (size, analysts, dispersion, prior return, book-to-market), FF12 sector effects, two-way clustering, Fama–MacBeth	**Done**; benchmark version run (§6.3)
2×2 table	`two_by_two.py`: beat/miss $\times$ good/bad tone (within-season median), cell means, SEs, 95% CIs, n, contrasts incl. diff-in-diff	**Done**
Text beyond surprise	Spec (3) coefficient, Clark–West, 2×2 within-row effects	**Done**
Robustness	`robustness.py`: sectors, size, NYSE, periods, Llama cutoff, all windows, placebo, stricter samples, dispersion-scaled SUE	**Done**
2022 rate shock	`rate_shock.py`: FOMC-dated regimes, per-regime IC and coefficient, pooled interaction + Wald test	**Done**; **Locked** (holdout)
Save outputs automatically	`outputs.py`: `.csv/.md/.tex`, provenance log (sha256, git commit, upstream versions), refuses row-level tables	**Done**

Quality checks

- **29 tests** (`python code/tests/run_all.py`): hand-computed answers (ranks, Newey–West equal to part 4's, Clark–West, exact 2×2 and beat/miss cell means, LM negation, folds without look-ahead, holdout lock, book-to-market timing and units, git provenance) and recovery tests on synthetic data (planted effect found; noise and placebo not; Clark–West size under the null over 15 seeds).
- **Notebooks**: `00_inputs_check`, `07_baselines`, `08_evaluation`, `08b_rate_shock`, each with markdown explaining the choices.

6 Results so far

6.1 Real data: the panel

Split	Events	With text	Seasons	First	Last
dev	35,508	105	44	2006-01-04	2016-12-22
validation	11,375	10	12	2017-01-04	2019-12-20

Counts match part 4's `sample_by_split.csv`. Only 115 events have text so far.

6.2 Real data: the surprise benchmark (validation, 2017–19)

Outcome	Rank IC	t_{NW}	Long-short	t_{NW}	AUC
Reaction, `car_0_1_ff4`	0.314	22.3	7.52%	13.3	0.655
`post_car_1_ff4`	0.026	2.5	0.22%	0.6	0.514
Drift, `post_car_60_ff4`	−0.011	−0.7	−1.35%	−1.9	0.497
`car_1_60_ff4`	−0.020	−1.3	−1.66%	−2.4	0.495
Placebo, `ar_m1`	0.036	4.7	0.26%	4.5	0.521

Reading: the surprise is strongly priced at release and shows no post-announcement drift in 2017–19. It also predicts the day −1 return, consistent with pre-announcement drift (leakage or anticipation). That is why the placebo is reported *controlling for SUE*: a text signal only fails the placebo if it predicts day −1 beyond what the surprise does. The out-of-sample R^2 of the surprise-only decile model on the reaction is 8.7%: this is the bar text models have to raise.

6.3 Real data: the benchmark regression

Reaction `car_0_1_ff4` (winsorised) on the surprise, two-way clustered by firm and season (`results/tables/cd_surprise_regressions__*`). This is specification (4) of the ladder without the text term: the regression every text signal will be added to.

	Dev 2006–16		Validation 2017–19	
	SUE rank	+ controls	SUE rank	+ controls
SUE rank	0.0830 (29.0)	0.0835 (29.4)	0.0789 (18.3)	0.0802 (19.4)
log market cap		−0.0012 (−3.6)		−0.0000 (−0.1)
log book-to-market		−0.0022 (−3.6)		−0.0019 (−1.8)
B/M missing or negative		0.0023 (0.8)		0.0107 (3.1)
Prior 1-month return		−0.0098 (−2.0)		−0.0060 (−0.5)
Dispersion / price		−0.41 (−2.4)		0.44 (1.6)
Sector FE	no	yes	no	yes
N	35,508	35,508	11,375	11,375
R^2	0.118	0.119	0.099	0.102

All specifications include season fixed effects; t-statistics in parentheses. Moving from the bottom to the top of the surprise distribution adds about 8 percentage points of two-day abnormal return. Controls add almost nothing ($\Delta R^2 \leq 0.003$): the surprise is the whole benchmark. Fama–MacBeth gives the same SUE coefficient (0.078, $t_{NW} = 20.3$ in validation). The controls matter for a different reason: a text signal correlated with size or value could otherwise look informative.

6.4 Real data: beat, meet, miss

The one-axis version of the 2×2 table (`cd_surprise_sign__*`); 95% CIs clustered by season.

Validation 2017–19	n	Share	Mean $CAR_{0,1}$	95% CI
Beat	7,997	70.3%	+1.31%	[1.04, 1.57]
Meet	559	4.9%	−1.48%	[−2.12, −0.83]
Miss	2,819	24.8%	−3.22%	[−3.56, −2.88]
Beat − miss			4.53%	[4.16, 4.89]

Dev years give the same picture (beat +1.67%, miss −3.29%). Misses are punished more than beats are rewarded, and a meet is bad news. When text signals arrive, the 2×2 table splits each of these rows by text tone; the question is whether the split is large.

6.5 Real data: is the benchmark stable?

Rank IC of SUE with the reaction, validation (`cd_surprise_robustness__validation`):

- **Sectors:** 0.18 (utilities, energy) to 0.41 (shops, finance); $t \geq 3.6$ everywhere estimable. Durables and telecom have too few events per season.
- **Size terciles:** 0.32 / 0.32 / 0.29 (small / mid / large); slightly weaker for large firms.
- **Years:** 0.32 (2017), 0.29 (2018), 0.33 (2019). **Sessions:** 0.30 after the close, 0.33 before the open.
- **Samples:** unchanged without `basis_suspect` events (0.315). `timing_suspect` is never set in the primary sample, so that check is empty there.

Sector heterogeneity is the one notable pattern: where SUE is weak (utilities, energy), text has more room to add information, so sector-level text results will be worth reporting.

6.6 Real data: the day −1 placebo

The surprise ranks the day −1 return in every subset (`cd_placebo_diagnostic`):

Split	Subset	IC with AR_{-1}	t_{NW}
Dev	all	0.052	9.1
	after close / before open	0.061 / 0.044	7.0 / 5.6
	small / mid / large	0.065 / 0.036 / 0.049	9.4 / 4.7 / 5.3
	2006–10 / 2011–16	0.052 / 0.051	5.0 / 8.3
Validation	all	0.036	4.7
	after close / before open	0.037 / 0.033	7.0 / 1.8

Reading. Every event in the primary sample has the wire and I/B/E/S clocks agreeing, so day 0 is unlikely to be one day late. For an after-close release, day −1 is the release day itself, whose close-to-close return ends before the release. The effect is about one-sixth of the reaction IC, diffuse across sessions, sizes and periods, and smaller in later years. That pattern fits pre-announcement drift (anticipation, information leakage, guidance) better than a systematic clock error. Parts 1 and 3–4 should still look at it: a subset of misdated releases would produce the same signature. For parts 7–8 the consequence is that the text placebo must control for SUE, which the robustness grid does.

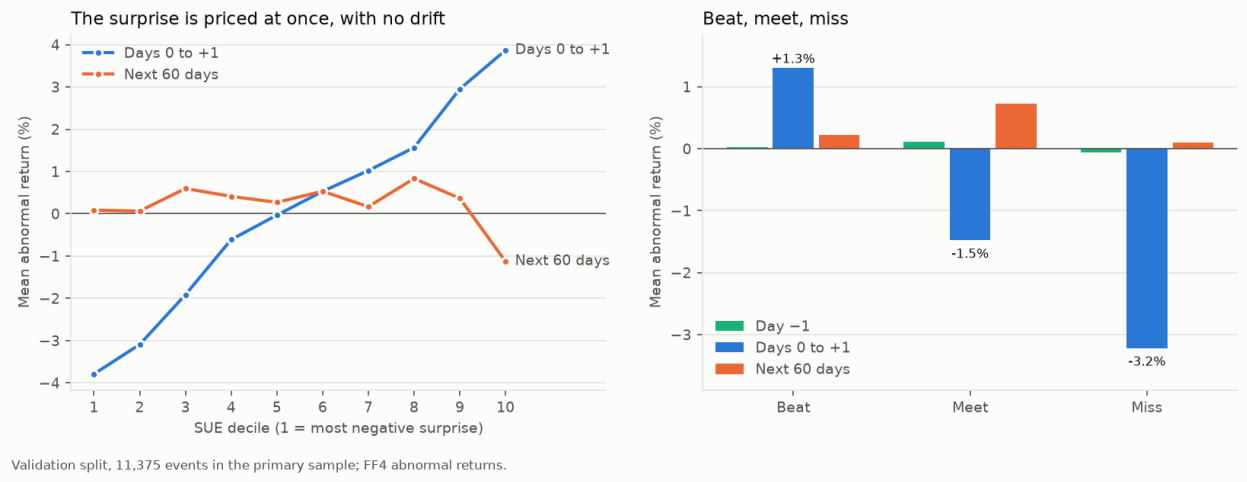
6.7 Real data: Loughran–McDonald pilot (exploratory)

Only 115 events have text (105 dev, almost all 2006; 10 validation), below the 5% coverage rule, so the pipeline does not evaluate LM. Purely to check direction (`cd_lm_pilot_*`):

- LM tone vs reaction: Spearman 0.16 ($p = 0.09$); vs SUE: 0.04, so nearly orthogonal to the surprise.
- Reaction on SUE rank + standardised tone (HC1): tone +0.73% per SD ($t = 1.6$), R^2 0.231 → 0.248; unchanged with size and B/M.
- 2×2 cells, tone above / below median: beat 2.1% / 0.4%; miss −1.2% ($n = 5$) / −4.7% ($n = 18$).

The direction matches the hypothesis (tone adds to the surprise in both rows), but none of this is a result: one year, $n = 115$, no seasons to cluster on. It shows the LM path works end to end on real texts.

6.8 Real data: one figure



Left: the reaction rises steeply with the SUE decile while the next 60 days are flat (the top decile even gives some back). Right: misses are punished about 2.5 times more than beats are rewarded, and a meet is treated as a miss. (`results/figures/cd_surprise_response__validation.png`.)

6.9 Real data: where the surprise gets it wrong

The surprise ranks reactions well on average, but event by event it often gets the *sign* wrong (`cd_surprise_errors__validation`):

Validation 2017–19	n	Reaction has the wrong sign	Wrong by more than 5%
All beats	7,997	41.3%	14.0%
All misses	2,819	32.9%	9.7%
Top SUE quintile	2,163	30.8%	11.8%
Bottom SUE quintile	2,164	32.5%	10.1%

Even in the most extreme surprise quintiles, nearly one event in three moves the other way. That residual is the room text has to work with.

Worked example. Conagra Brands, results released before the open on 20 December 2018: EPS 0.67 against a median consensus of 0.56 (10 analysts), a top-quintile surprise (SUE rank 0.40). The surprise predicts a gain; the two-day abnormal return was −23.0%, and a further −29.2% over the next 60 trading days. Public reporting attributes the drop to a cut in full-year guidance after the Pinnacle Foods acquisition. This is the error type the project is about (a guidance revision that the EPS number cannot carry), and it continued to drift, so the market did not fully price it on day 0–1. *To verify against the 8-K text (accession 0001193125-18-353858) once part 5's texts are available.* The list of the 20 largest disagreements among large firms is kept in `data/error_cases__validation.csv` (private: it names firms).

6.10 Sample sizes and power

Counts from part 3–4's event table (`cd_sample_counts`; holdout outcomes are not read):

Split	Years	All events	Text sample	Primary	Primary, post-cutoff	Quarters
dev	2006–16	120,439	51,529	35,508	0	44
validation	2017–19	30,647	14,523	11,375	0	12
excluded	2020	10,469	5,232	4,114	0	4
holdout	2021–25	56,056	26,503	22,145	8,579	20
total		217,611	97,787	73,142	8,579	

"Post-cutoff" is after Llama 3.1's December 2023 training cutoff. These are part 3–4's `sample_v1` counts (6 November export); the checkpoint funnel (115,561 events) uses a different filter set, and the two should be reconciled before they are reported side by side.

Minimum detectable incremental IC (`cd_mde_incremental_ic`; two-sided 5%, 80% power, quarters as the independent units, t_{S-1} critical values). The quarterly IC standard deviation is estimated two ways: a random signal on the real validation returns (best case, 0.032), and the observed variation of the SUE IC across quarters (realistic for a signal whose strength varies, 0.059).

	Events	Quarters	MDE, best case	MDE, realistic
Validation	11,375	12	0.028	0.052
Holdout	22,145	20	0.021	0.039

Implication. The pre-registered bar (incremental IC ≥ 0.02) is at the edge of what the holdout can detect in the best case, and below it in the realistic case. Two caveats cut the other way: an incremental IC is a *paired* difference (SUE+text minus SUE on the same events), whose quarterly variance is lower than a single IC's, so these numbers are conservative; and the regression tests pool events rather than quarters. Still, a true incremental IC of 0.02 could be missed, and the report should say so. Options to discuss: pre-register the paired test explicitly, pool validation and holdout for the main test (with model selection frozen on dev), or treat 0.02 as the economic threshold and report confidence intervals rather than a pass/fail.

Contamination comparison (`cd_mde_contamination`), with the checkpoint's formula $SE \simeq \sqrt{2(1-r)(1/n_a + 1/n_b)}$ and the actual holdout arms (13,566 pre-cutoff vs 8,579 post-cutoff primary events): MDE = 0.039, 0.030 and 0.017 at $r = 0.5$, 0.7 and 0.9. The checkpoint's example (0.061 at 2,500 per arm, $r = 0.7$) is reproduced exactly; the real sample is about twice as sensitive.

6.11 Synthetic data: the pipeline works

Synthetic events carry a planted text effect (a "good" LLM) and a pure-noise LLM. Combined model vs surprise-only, reaction window, validation split:

Text signal	Text-only IC	OOS R^2 gain	Clark–West t	p
Good LLM	0.314	+3.8 pp	17.1	$< 10^{-8}$
LM tone	0.295	+3.0 pp	10.2	$< 10^{-6}$
TF-IDF ridge	0.259	+3.0 pp	6.0	$< 10^{-4}$
Noise LLM	0.010	+0.0 pp	0.9	0.19

These numbers are not findings; they show the machinery finds an effect that exists and does not invent one that does not.

7 What is missing

7.1 Upstream inputs (blocking)

Item	Who	Effect on 7–8	Action here
Release texts 2007–2025	Part 5	LM and TF-IDF cover 115 events; TF-IDF needs ≥ 500 training texts per fold; no real regression or 2×2 with text	Rerun `--step all`
LLM scores	Part 6	No LLM rows in any table	Rerun `--step evaluate`
Corrected release clock	Part 1, then 3–4	A WRDS rerun with combined acceptance times is on `main` (8 Nov); our numbers use the 6 Nov event table. Returns and surprise may shift; may also explain part of the day -1 placebo	Refresh `events_final`, rerun `--step all`
Holdout freeze	Part 5 (`splits.py --freeze`)	Test run and rate shock locked	Final run (§8)

7.2 Decisions to write down before the freeze

These must be fixed in `docs/METHODOLOGY.md` (ideally agreed with the instructor) before the holdout is opened; afterwards they cannot change.

1. **Headline outcome.** Proposed: `car_0_1_ff4` (reaction) as primary; the 60-day post-release window as secondary.
2. **Main sample.** `in_primary` (3+ analysts) vs the broader sample; with or without the `basis_suspect` and `timing_suspect` events.
3. **Main LLM signal.** Which model, prompt and variant (`raw` vs `mask`) is the headline in the 2×2 table and the regressions. The others become robustness rows.
4. **Size filter.** Whether to drop micro-caps (e.g. below the NYSE 20th percentile).
5. **Multiple testing.** How many signals $\times$ windows are "primary"; the rest are labelled exploratory.
6. **Interpretation table.** What each pattern of results would mean (priced in vs under-reaction vs memorisation), written before seeing the holdout.

7.3 Extensions not yet built (own work, optional)

- **Tradable portfolio.** Current long-short uses same-quarter ranks (fine for measuring information, not a strategy). A version with breakpoints from trailing releases only would give an implementable return.
- **FactSet consensus robustness.** If part 3 provides it, add an alternative `sue_*` column to the robustness grid.

- **Figures for the report.** Cumulative abnormal return by text tercile within beat/miss (event-time plot), IC by season over time.

8 How to continue

All commands run from the folder `7_8_baselines_evaluation/`.

1. **Today's state:** `python code/pipeline.py --step all`, then `python code/current_data.py` for the benchmark, placebo and pilot tables (`cd_*`). Book-to-market is cached in `data/`; `python code/book_to_market.py --refresh` re-pulls Compustat (needs a Duo approval).
2. **When part 5 delivers more texts:** rerun `--step all`. Sanity-check the TF-IDF top terms (they should be words like "impairment", "raised guidance", not firm names), text coverage by year, and the placebo row.
3. **When part 6 fills the model cache:** `python code/pipeline.py --step evaluate`. LLM signals appear automatically in every table.
4. **Make the decisions above** and record them in `docs/METHODOLOGY.md` and `docs/CHANGELOG.md`.
5. **After the freeze, once:**
   `UNLOCK_HOLDOUT=1 python code/pipeline.py --step all --eval-split holdout`, then run `08b_rate_shock.ipynb`.
6. **Hand over to part 10:** the `.tex` tables in `results/tables/` can be `\input` directly into the report.

What each upstream change requires

Upstream change	Rerun
Part 5 cleaned more texts	`07_baselines`, then `08_evaluation` (LM counts are cached)
Part 6 added model scores	`08_evaluation` only
Part 1/3/4 corrected clock, surprise or returns	Everything (`--step all`)
Holdout frozen	The single final run

A File map

File	Role
`code/settings.py`	Paths and research-design constants (outcomes, splits, dates, seeds)
`code/io_inputs.py`	Readers for parts 3–4, 5, 6 with column contracts
`code/panel.py`	Analysis panel: seasons, ranks, controls, winsorising, holdout guard
`code/crossfit.py`	Chronological out-of-sample fitting with embargo
`code/baseline_surprise.py`	Benchmark 1: SUE
`code/baseline_lm.py`	Benchmark 2: Loughran–McDonald
`code/baseline_tfidf.py`	Benchmark 3: TF-IDF + ridge / logistic
`code/metrics.py`	IC, long-short, OOS R^2, Clark–West, hit rate, AUC
`code/compare.py`	Surprise-only vs text-only vs combined
`code/regressions.py`	Ladder, two-way clustering, Fama–MacBeth
`code/two_by_two.py`	2×2 table and contrasts
`code/robustness.py`	Robustness grid
`code/rate_shock.py`	2022 rate-shock regimes
`code/book_to_market.py`	Compustat book equity as of the release, divided by market cap
`code/current_data.py`	Benchmark regression, beat/miss, stability, placebo diagnostic, LM pilot, error cases, sample counts, power, figure
`code/outputs.py`	Saving and provenance logging
`code/synthetic.py`	Synthetic data with planted effect
`code/pipeline.py`	Command-line entry point
`docs/METHODOLOGY.md`	Why everything is done this way
`docs/INTERFACES.md`	Column and score contracts with other parts

Data note. This report contains summary statistics only. Row-level WRDS derivatives stay in `data/` on the team Drive and are never committed or shared.